

Week 1 Worksheet: Introduction to Quantum Computing

Cornell Quantum Computing Association

Name: _____

Date: _____

Overview Questions

1. What are the two fundamental states that form the computational basis in quantum computing?
2. In your own words, explain what makes a quantum coin different from a classical coin.
3. What happens when you measure a qubit that's in superposition?

Classical Foundation

4. If you have 4 classical coins, how many total arrangements are possible?
5. Write the vector representation for a classical bit in state 1:
6. True or False: Classical probability (like "50% heads, 50% tails") represents genuine uncertainty about the physical state of the system.

Quantum Concepts

7. Complete the quantum state notation. If a classical bit in state 0 is written as $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, then in quantum notation this is: _____

8. A qubit is in the state $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$.

- (a) What is the probability of measuring this qubit in state $|0\rangle$?
- (b) What is the probability of measuring this qubit in state $|1\rangle$?
- (c) Verify that these probabilities sum to 1.

9. Multiple choice: The normalization condition $|\alpha|^2 + |\beta|^2 = 1$ ensures that:

- (a) The quantum state has the right energy
- (b) The measurement probabilities sum to 1
- (c) The qubit rotates properly
- (d) The quantum computer runs faster

10. What is the key mechanism that allows quantum algorithms to provide computational advantage despite measurement collapse?

Synthesis and Extension

11. Consider a problem where you need to find one specific arrangement out of 2^{20} possible arrangements.

- (a) How many steps would a classical computer need in the worst case?
- (b) How many qubits would you need to represent all possibilities in superposition?

- (c) Why can't you just put the qubits in superposition, measure them, and immediately get the right answer?

12. Thought experiment: Imagine you have a quantum coin that's in the superposition state $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$.

- (a) Is this a valid quantum state? Show your work by checking the normalization condition.
- (b) If you measured this coin 300 times, approximately how many times would you expect to get outcome 0?
- (c) How is this different from a classical coin that lands heads with probability $\frac{1}{3}$?

Looking Ahead

- 13.** What matrix represents the quantum NOT gate (X gate)?
- 14.** If qubits can be represented as vectors, what can quantum gates be represented as?